

ngspice-46 — Full Change Report

Every modification and its reason (original → current)

Ngspice + OpenVAF Enhancements project

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ngspice-46 — Full Change Report

Every modification applied to ngspice-46 in this project, with its reason. The baseline is the pristine ngspice-46 source tree (as vendored in the project's `original/` snapshot, which already contains OpenVAF-Reloaded's stock OSDI support); the current state is the tree committed in this repository under `ngspice-46/`. Each entry links the enhancement write-up in [enhancements_doc/](#) that carries the full engineering detail and the verifying example suite. The companion document [openvaf_changes_full-report.md](#) covers the compiler side.

Maintenance note: this report is updated whenever an enhancement touches ngspice sources. If you are reading it alongside a newer tree, the per-enhancement index at the end tells you the last change it covers.

Scope summary. One new source file and 41 modified ones carry all the functional changes (~2,000 substantive diff lines). A further ~100 `Makefile.in` files and `config.h.in` differ only because the auto-tools were regenerated with a current libtool ([E-77](#)); they contain no hand-written changes and are not listed individually. Everything below is behavior, interface, or diagnostics.

1. The OSDI ABI, version 0.4 → 0.7

`src/osdi/osdi.h` is the authoritative C contract between ngspice and the `.osdi` objects `openvaf-r` produces. The project extended it four times — twice additively (new exported symbols old loaders simply never look up), twice with genuine version bumps (layout changes):

Change	Kind	Enhancement	Why
<code>OSDI_ABSDELAY_COUNT</code> / <code>OSDI_ABSDELAY_INFO</code> exports	additive sym- bols	E-1	<code>absdelay()</code> needs simulator-side waveform history and synthetic delay-equation rows; the descriptors tell ngspice which nodes/offsets each delay slot uses
<code>OSDI_LAST_CROSSING_COUNT</code> / <code>OSDI_LAST_CROSSING_INFO</code> exports	additive sym- bols	E-6	<code>last_crossing()</code> needs waveform history and crossing interpolation — same additive pattern as <code>absdelay</code>
<code>OsdiNode.nodeset</code> field	ABI 0.5 (node stride change)	E-45	Verilog-A net initializers (<code>electrical a = 5.0;</code>) become solver initial-guess nodesets; every node record carries the hint
<code>num_ac_stim_src</code> / <code>ac_stim_sources</code> / <code>load_ac_stim</code> descriptor tail	ABI 0.6 (de- scriptor grows)	E-51	full <code>ac_stim()</code> support: a Verilog-A module can <i>be</i> the AC stimulus, so the descriptor must enumerate its AC right-hand-side sources
<code>load_noise</code> destination stride 1 → 2 (signed (<code>flat</code> , <code>react</code>) power pairs)	ABI 0.7	E-54	correct noise physics: operating-point-dependent factors and <code>ddt()</code> -shaped ($j\omega$) noise need a complex amplitude per source, $re + j\omega \cdot im$, so coherent same-named sources (LRM 4.6.4) sum with correct sign and frequency shape

Two additive flag conventions ride on the existing `eval()` interface (not version bumps):

- `EVAL_FLAG_IS_FINAL_STEP` (bit 1 ≪ 21) in the `eval input` flags — `@(final_step)` firing ([E-53](#));
- `eval return` flags for `$finish/$stop` requests and `$discontinuity`-driven step rejection ([E-55](#)).

Pairing consequence: this ngspice requires $\text{OSDI} \geq 0.7$ and rejects older objects with a clear version message; `.osdi` files from older compilers must be recompiled (E-54).

2. The OSDI runtime — `src/osdi/`

osdiaccept.c — new file (E-55, E-1, E-6)

The accepted-timepoint hook (`DEVaccept`) OSDI previously lacked. It advances the `absdelay/last_crossing` waveform histories only at *accepted* points (so rejected timesteps don't corrupt the delay lines) and reads the per-attempt latched `$finish/$stop` request flags at the one boundary where honoring them is safe. Why: acting on `$stop` mid-Newton-iteration was treated as a step failure and ground the timestep into a rejection loop; `$finish` was ignored outright.

osdidefs.h (104 diff lines)

- `OsdiExtraInstData` grows the per-instance fields behind every runtime feature: `dt/temp` offsets with given-flags (instance temperature), `eval_flags`, `absdelay` waveform-history rows and pre-allocated KLU/SPARSE matrix-pointer sets for the delay-equation rows (re-pointed on every DC↔AC transition, mirroring the regular Jacobian handling), and the `last_crossing` history (E-1, E-6, E-55).
- `ALIGN` macro renamed `OSDI_ALIGN`: the macOS SDK's `<sys/param.h>` defines an unrelated `ALIGN`, producing a redefinition warning in every OSDI translation unit (E-77).

osdiitf.h / osdiext.h

The registry-entry structure gains the `absdelay/last_crossing` descriptor pointers, and `OSDIpendingRequests()` is exported so the analyses can ask "did any device request `$finish/$stop` at this accepted point?" (E-1, E-6, E-55).

osdiregistry.c (68 diff lines)

- Loads the `absdelay/last_crossing` descriptor arrays from each `.osdi` and threads them into the registry entries (E-1, E-6).
- Requires $\text{OSDI} \geq 0.7$ (E-54).

osdiinit.c (8 diff lines)

- Registers the `DEVaccept` hook (E-55).
- Publishes the synthetic instance-temperature parameter under both spellings, `dt` and the conventional `dtemp` every built-in device uses; `n1 a b mod dtemp=10` used to fail with "*unknown parameter*" while the underlying plumbing was exact (E-80).

osdiload.c (441 diff lines — the largest OSDI change)

- Stamps the `absdelay` synthetic rows (`y_synth`, `z`) for DC, AC and transient, driving the delay from the recorded history (E-1).
- Evaluates `last_crossing` from the recorded waveform with linear interpolation between the bracketing timepoints (E-6).
- Passes the final-step flag through to `eval()` (E-53).
- Latches `eval`-return flags per timepoint attempt (`point_eval_flags`, OR-ed across Newton iterations, reset per attempt) so `$finish/$stop` under solution-dependent conditions survive to the accepted-point boundary (E-55).

- Folds the stride-2 signed noise-power pairs ($\text{fac} \cdot |\text{fac}|$ semantics) when transferring `load_noise` results (E-54, E-42).

osdisetup.c (178 diff lines)

- Creates the synthetic delay-equation unknowns and matrix entries for `absdelay` (E-1).
- Applies the OSDI nodesets (`OsdNode.nodeset`) as solver initial guesses (E-45).
- A `$fatal/$finish` raised during setup (models rejecting their configuration by design — HiSIM's `$port_connected` guards) used to surface as ngspice's baffling *"impossible error — can't occur"*; it now reads *"a Verilog-A device rejected its configuration during setup"* (E-56).

osdiacld.c (89 diff lines)

- AC loading gains the `ac_stim` right-hand-side injection: the partitioned AC-stimulus source array is loaded through `load_ac_stim` and added to the AC RHS with exact magnitude and phase, `$mfactor` applied linearly (deterministic signals scale with multiplicity) (E-51, E-26).
- Complex ($j\omega$ -shaped) noise coupling entries participate in the AC matrix (E-54).

osdinoise.c (93 diff lines)

- Per-instance grouping of same-named noise sources: coherent summation of complex amplitudes ($a + j\omega \cdot b \cdot T$ before squaring, per LRM 4.6.4 — same-named sources correlate (including exact anti-phase cancellation), differently-named ones stay independent (E-42).
- Reads the stride-2 signed pairs of ABI 0.7 (E-54).

osditrunc.c (34 diff lines)

- Honors `$discontinuity(n)`'s next-step clamp: a negative `bound_step` sentinel from the model limits the step after the event (E-24).
- Honors the step-rejection return flag: when an event fired inside the step, `OSDIt trunc` requests `delta/8` (with a $20 \cdot \text{CKTdelmin}$ termination floor) so the integrator bisects onto the discontinuity instead of extrapolating across it (E-55).

osdi/Makefile.am

Adds `osdiaccept.c` to the build (E-55).

3. Analyses — **src/spicelib/analysis/**

dctran.c (46 diff lines)

- Advances OSDI delay/crossing histories via the accept path and skips history corruption on rejected steps (E-1, E-6).
- Issues the dedicated `@(final_step)` evaluation at the converged end of a successful transient (E-53).
- Honors accepted-point `$finish` (ends the analysis cleanly, firing `final_step` at the requesting point), `$stop` (pauses resumably), and aborts with a clear error on `$fatal`'s `E_PANIC` instead of retrying it as nonconvergence (E-55).

dcop.c (9) and **acan.c** (10)

- Fire `@(final_step)` at their successful end (an operating point fires both step events — a single point is first *and* last) (E-53).

- An AC job's operating-point phase carries the analysis name bit (only the name — adding the re-active CALC_* bits would wrongly enable integration at an op), so analysis("ac") holds through the whole AC run per LRM 4.6.1 and @(initial_step("ac")) fires in AC (E-53).

dctrcurv.c (150) + **include/ngspice/trcvdefs.h** (13)

- Generic **.dc @inst[param]** sweeps (a new sweep code): the DC sweep previously hardcoded Vsource/Isource/Resistor/temperature, so sweeping any other device parameter was a fatal error. The generic sweep resolves @inst[param] through the device's own parameter tables, refreshes per point via DEVtemperature (for OSDI exactly the alter path), nests with other sweep variables, and restores the original value afterwards (E-62).
- Fires final_step at the last sweep point; honors a \$finish raised mid-sweep (E-53, E-55).

noisean.c (37)

- A singular AC matrix during noise analysis crashed ngspice with a SIGABRT: the code ignored NIacIter's return and the adjoint solve asserted on the unfactored matrix. It now aborts cleanly ("AC solution failed at ... Hz") (E-56).
- Honors deferred \$finish/\$stop raised at the noise operating point and fires final_step on success (E-55, E-53).

span.c (31) + **cktspdum.c** (14)

- 1-port S-parameter analyses enabled: the "we need at least two ports" error was over-strict (a 1-port is a plain reflection measurement; the matrix machinery is N-general) — and it was hiding the cadjoint crash fixed in dense.c (E-64).
- **Rbase** published into the sp plot from port 1's z0, making the Touchstone writers work with no manual let Rbase = 50 incantation (E-64).
- Stock NaN fixed in **donoise** noise-parameter extraction: the uncorrelated noise conductance Gu is analytically zero for fully-correlated single-source topologies, and floating-point rounding could land sqrt(Ycor² + Gu/Rn)'s argument at $-10^{-18} \rightarrow \text{NaN}$ for the noise figure. Clamped to the physical range ≥ 0 — found by parity testing against an OSDI resistor that produced the exact textbook NF where the built-in returned NaN (E-63).

cktdisto.c (16)

- **.disto** silently reported zero distortion for OSDI devices — the distortion kernel needs higher-order Taylor coefficients the OSDI ABI cannot provide (first derivatives only), and such devices were skipped without a word. A prominent warning now names each affected OSDI device type (E-62).

4. Frontend — **src/frontend/**

plotting/pyplot.c (new) + **com_pyplot.c** (new) + **plotit.c**, **commands.c** (E-94)

A new interactive command, **pyplot**, plots simulated vectors with matplotlib — a Python counterpart to gnuplot. ft_pyplot() (plotting/pyplot.c) mirrors ft_gnuplot(): it writes a <file>.data table and a <file>.py matplotlib script and system()'s python3 (synchronously for a PNG, backgrounded for an interactive window). A new "pyplot" device arm in plotit() routes to it, so it accepts the same plot expressions as plot/gnuplot; the com_pyplot() wrapper mirrors com_gnuplot(), and pyplot is registered in both command tables. Two set variables tune it: pyplot_terminal=png renders headless (Agg) to a PNG, and pyplot_python picks the interpreter (default python3). Purely additive (E-94). The output file base name is optional (E-95): com_pyplot() treats the first word as a file name only

when it is not a plot expression (no `(`, and not a vector by `vec_get()`), otherwise it defaults to `pyplot` — so `pyplot v(out)` plots directly (the command's minimum argument count was lowered from 2 to 1). `ft_pyplot()` also grew stacked subplots (set `pyplot_subplots=N` → `plt.subplots(nrows, 1, sharex=True)`), `N` traces per panel; `vs` stays the x-axis, so it is not a panel separator) and matplotlib style sheets (set `pyplot_style=<name>`, `dark` → `dark_background`, guarded) (E-98). The headless `pyplot_terminal` was extended from PNG-only to the vector export formats `svg` and `pdf` (set `pyplot_terminal=svg/pdf` → `fig.savefig(<file>.<fmt>)`, matplotlib picking the writer from the extension), and a figure size control was added (set `pyplot_figsize="W,H"` → `plt.subplots(..., figsize=(W, H))`; quote the value so ngspice keeps the comma) (E-99).

postcoms.c (414 diff lines) + **postcoms.h** + **commands.c** (16)

The Touchstone I/O subsystem (E-64, E-72):

- **wrsnp** (with `wrs2p` dispatching to the same handler): Touchstone v1 for any port count — the classic 2-port `S11 S21 S12 S22` column order preserved byte-identically, $N \geq 3$ in the spec's row-major layout with at most four complex pairs per line;
- writer options `wrsnp <file> [ri|ma|db] [s|y|z] [hz|khz|mhz|ghz]`, any combination in any order: MA/DB formats, Y/Z parameters (normalized to `Rbase`, $Y \cdot R / Z / R$, per the v1 spec), frequency units — the option line reflects every choice;
- **rdsnp**: reads any Touchstone v1 file into a new plot with a Hz frequency scale and complex vectors matching the `.sp` plot's conventions (MA/DB converted back, Y/Z de-normalized, the 2-port column order handled, `Rbase` published so imports round-trip) — measured data diffs against simulation in one `let` expression.

Why: E-63 showed the S-parameter *analysis* was exact but its results could not leave ngspice in the industry-standard format (`wrs2p` demanded a never-created `Rbase` vector), and nothing could bring VNA data in.

outitf.c (16)

- Integer operating-point variables recorded per point: `getSpecial` masked with `IF_REAL` only, so an integer opvar saved with `.save @n1[flag]` produced garbage in transient vectors (E-32).
- The plot-memory guard's message printed `%Id` — a Windows-only printf length modifier that is undefined behavior elsewhere and rendered the message as the numberless "memory required (Id Bytes)". Now `%zu`: real byte counts on every platform (E-77).

resource.c (27)

`ft_ckptspace()` — the "approaching max data size" warning (`RSS > 95%` of `RSS + free RAM`) — now honors **set no_mem_check** (the same opt-out the fatal output-size estimate in `outitf.c` respects; one variable governs both memory guards) and warns once per excursion above the threshold, re-arming below 90%, instead of repeating on every check through a long analysis (E-81).

display.c (1)

`#undef NODEV` before the local macro definition — the macOS SDK's `<sys/param.h>` defines an unrelated `NODEV` (E-77).

5. Parser and device registry — `src/spicelib/`

`parser/ingtok.c` (10)

A `.model` card override silently dropped its first parameter when the type name and the opening parenthesis had no space between them (`modname devtype(p1=v1 ...)`): `INPgetNetTok`'s scan did not treat `(` as a token boundary, so the type token swallowed the paren and the first parameter name. Found while verifying event-function counters whose model-card overrides mysteriously kept defaults (E-8).

`parser/ingmod.c` (5)

`find_model_parameter()` dereferenced `*(device->numModelParms)`, which is `NULL` for built-ins that take no model cards (`VCVS`, `CCCS`, ...) — any *referenced* `.model m vcv()` card segfaulted, with no OSDI involved at all (an ordinary MOS instance sufficed). This was the true root of the long-documented "module named like a built-in crashes" gotcha. One `NULL` guard; both shapes now produce clean, located errors (E-76).

`devices/dev.c` (51)

- Duplicate device-type registration warned and skipped: `osdi_add_device()` appended every descriptor unconditionally, and the model-card lookup scans front-to-back — so a module name duplicated across two loaded libraries silently resolved to the *first* registration (loading an updated model library gave the stale device with no hint), and a module named like a built-in corrupted model creation. Registration now checks the table case-insensitively and skips duplicates loudly (E-76).
- `pre_osdi` path dedup: loading an already-loaded file notes it and skips — with the remedy stated: *"restart ngspice to load a recompiled file"* (E-76, E-81).

`osdi/osdiparam.c` (E-93)

Setting a fixed (`localparam`) OSDI parameter from the netlist used to be swallowed silently: `openvaf` exports every `localparam` as a parameter, its "given" flag is forced false, and the written value is overwritten by the parameter-init default — so `.model ... N=8` on a frozen structural width parameter changed nothing, with no feedback. `OSDIparam/OSDImparam` now test the new `PARAM_FLAG_FIXED` descriptor flag (set by `openvaf`, a free bit of the parameter `flags` field) and emit *"parameter 'N' is a fixed (localparam) value and cannot be set from the netlist; ignored"* instead of silently dropping it (E-93).

6. Maths — `src/maths/`

`dense/dense.c` (11 substantive lines; the file also carries a whitespace/line-ending normalization from editing)

`cadjoint()` had no 1×1 base case: its cofactor loop allocated 0×0 and then negative-sized minors, killing `ngspice` with `malloc: can't allocate -8 bytes` — the crash that had been hiding behind the "at least two ports" guard in `span.c`. The adjugate of `[a]` is `[1]` (the empty minor's determinant is 1 by convention), making `cinverse` of a 1×1 equal $1/a$; a $100\ \Omega$ load on a $50\ \Omega$ port now writes a proper `.s1p` with $S_{11} = 1/3$ exactly (E-64).

`sparse/spfactor.c` (6)

The Markowitz-product overflow guards compare a `double` against `LARGEST_LONG_INTEGER` (`= LONG_MAX`), which is not exactly representable as a `double`; the conversion is now explicit — identical semantics, intentional and warning-free (E-77).

KLU/klu_multiply.c (16)

SMPmultiply's KLU path passed NULL internal↔external ordering maps to KLU_matrix_vector_multiply, which dereferenced them unconditionally (&NULL[n] = 0×14 for n = 5) and segfaulted — so .option linesearch crashed under .option klu. The line search is the only caller of SMPmultiply under KLU, so this path had never been exercised. A NULL map now means the identity ordering (ext = i + 1, since the KLU CSR is 0-based and the RHS/Solution vectors are 1-based), making the KLU matrix-vector product — and hence the E-111 line-search residual merit — correct under KLU, with a merit sequence numerically identical to Sparse 1.3 (E-112).

KLU/klusmp.c (SMPcaSolve)

SMPcaSolve is the complex adjoint (transposed) solve used by noise and S-parameters. Its Sparse branch calls spSolveTransposed, but the KLU branch called the *non-transposed* klu_z_solve — silently wrong for any asymmetric matrix (every circuit with a transistor or controlled source), which is why .noise was disabled under KLU. It now calls klu_z_tsolve (transposed), so KLU noise matches Sparse exactly (including OSDI device models). This is the core of E-113, which also removes the KLU refusal guards in noisean.c / pzan.c (single-ended pole-zero runs under KLU; balanced-output pole-zero keeps a targeted guard).

7. Build system

xspice/icm/makedefs.in (4)

The XSPICE codemodel (.cm) link rule hardcoded the pre-macOS-10.3 idiom -bundle -flat_namespace -undefined suppress, deprecated loudly by the modern linker on all 14 codemodel links (CI macOS builds included). Now -bundle -undefined dynamic_lookup — the modern spelling for plugins whose host symbols resolve at load time (E-77).

Autotools regeneration (the ~100 Makefile.in files + config.h.in)

Regenerated by ./autogen.sh with libtool 2.5.4 during the zero-warning work — a build tree whose configure predates libtool 2.4.7 selects the deprecated darwin -undefined suppress flag whenever MACOSX_DEPLOYMENT_TARGET is unset. No hand-written content (E-77).

8. Per-enhancement index

Every enhancement that touched ngspice, oldest first:

Enhancement	ngspice files	One line
E-1	osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c, osdisetup.c, dctran.c (+accept path)	absdelay(): synthetic delay rows + waveform history
E-6	osdi.h, osdidefs.h, osdiitf.h, osdiregistry.c, osdiload.c	last_crossing(): history + interpolated crossings
E-8	parser/ingptok.c	.model card first-parameter drop fix
E-24	osditrunc.c	\$discontinuity next-step clamp
E-25	osdi callbacks (get_simpams)	expose analysis_name + simulator simpams
E-32	outitf.c	integer opvars recorded per point

Enhancement	new spice files	One line
E-42	osdinoise.c	coherent same-named noise grouping (LRM 4.6.4)
E-45	osdi.h (ABI 0.5), osdisetup.c	Verilog-A nodesets applied to the solver
E-51	osdi.h (ABI 0.6), osdiacld.c	full ac_stim AC-RHS injection
E-53	dctran.c, dcop.c, dctrcurv.c, acan.c, noisean.c, osdiload.c	@(final_step) + analysis-name phase bits
E-54	osdi.h (ABI 0.7), osdinoise.c, osdiload.c, osdiacld.c, osdiregistry.c	node-free complex noise factors, stride-2 pairs
E-55	osdiaccept.c (new), osdiload.c, osditrunc.c, osdiinit.c, dctran.c, dctrcurv.c, osdiitf.h, Makefile.am	\$finish/\$stop/\$fatal honored; \$discontinuity step rejection
E-56	noisean.c, osdisetup.c	noise SIGABRT fix; setup-rejection diagnostics
E-62	dctrcurv.c, trcvdefs.h, cktdisto.c	generic .dc @inst[param] sweeps; .disto warning
E-63	span.c	donoise NaN clamp (stock defect, found by parity)
E-64	span.c, cktspdum.c, dense.c, postcoms.c/.h, commands.c	Touchstone export, auto-Rbase, 1-port .sp, cadjoint 1x1
E-72	postcoms.c/.h, commands.c	Touchstone round 2: MA/DB/Y/Z/units + rdsnp reader
E-76	dev.c, inpgmod.c	duplicate-registration warning, load dedup, stock .model segfault fix
E-77	osdidefs.h, display.c, outitf.c, spfactor.c, makedefs.in (+autotools regen)	zero-warning build, 33 → 0
E-80	osdiinit.c	dtemp instance-parameter alias
E-81	resource.c, dev.c	memory-warning opt-out + once-per-excursion; reload-note hint
E-93	osdi.h, osdiparam.c	warn (not silently ignore) when a netlist sets a PARAM_FLAG_FIXED (localparam) OSDI parameter
E-94	plotting/pyplot.c (new), com_pyplot.c (new), plotit.c, commands.c	new pyplot command — plot vectors with matplotlib (a Python gnuplot)
E-95	com_pyplot.c, commands.c	make the pyplot output file name optional (defaults to pyplot)
E-98	plotting/pyplot.c	pyplot stacked subplots (pyplot_subplots=N) + matplotlib style sheets (pyplot_style)
E-99	plotting/pyplot.c	pyplot vector export formats (pyplot_terminal=svg/pdf) + figure size (pyplot_figsize)
E-110	optdefs.h, tsdefs.h, cktsopt.c, cktntask.c	.option errpreset=conservative moderate liberal — one knob for a coordinated tolerance/robustness set; explicit options override regardless of order (moderate = historical defaults)
E-111	optdefs.h, tsdefs.h, cktdefs.h, cktsopt.c, ckttojob.c, cktntask.c, cktdest.c, niiter.c	.option linesearch — globalized (damped) Newton via Armijo backtracking on a new KCL-residual merit □ F□ =□ G·x-b□ ; result-neutral, off by default

Enhancement	spice files	One line
E-112	maths/KLU/klu_multiply.c	KLU support for <code>.option linesearch</code> — SMPmultiply's KLU path passed NULL ordering maps that <code>klu_matrix_vector_multiply</code> dereferenced (SIGSEGV); NULL now means identity ordering. Line search verified merit-identical KLU vs Sparse
E-113	maths/KLU/klusmp.c, spicelib/analysis/noisean.c, pzan.c	KLU support for noise + single-ended pole-zero — SMPcaSolve's adjoint KLU branch used the non-transposed solve (silently wrong noise on asymmetric matrices); now <code>klu_z_tsolve</code> . Guards removed; balanced-output pz keeps a targeted guard

(E-25's `simpam` exposure lives in the OSDI callback table populated at load time; its diff rides inside the `osdi_load.c/callbacks` changes.)

Every entry above is guarded by a verify script under [examples/](#) and was regression-checked against the full example arsenal, the compiler's integration suite, and the VA_TEST industry-model corpus at the time it landed.